

HUMAN KNOWLEDGE PROTECTION SYSTEM

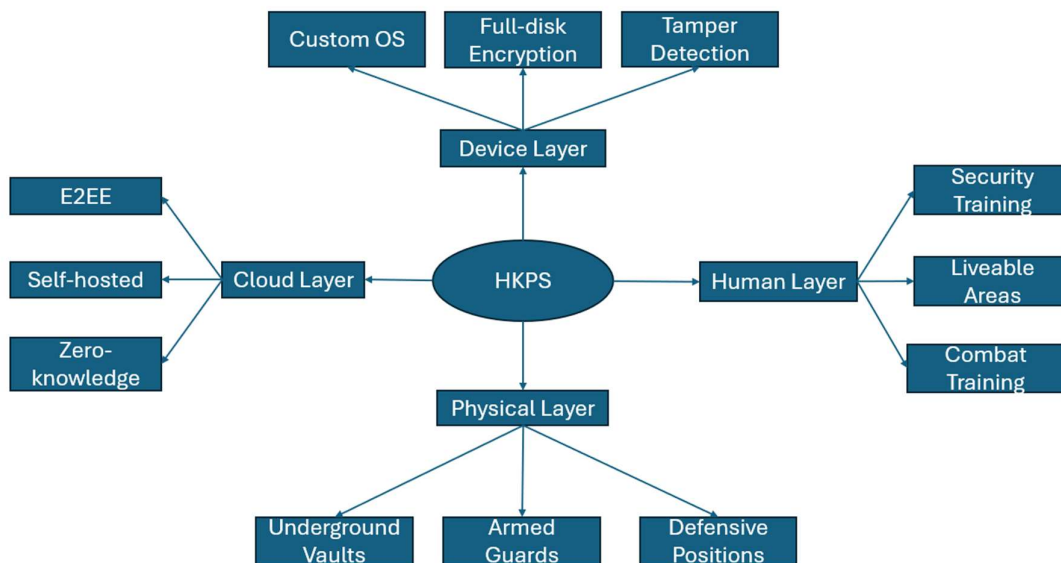
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Introduction

This human knowledge protection system otherwise known as HKPS was designed to ensure that humanity's knowledge survives the most extreme but realistic cases. It is also designed to ensure human life continues to persist even after events like nuclear winter. This system was developed as currently there are multiple threats to our world that could send humanity back decades. This human knowledge protection system document will explain the different layers & features; the workflow of how this system could be active daily and the trade-offs. It is important to understand that this system is designed to survive most threats meaning it will require large amounts of resources.

This human knowledge protection system cannot promise perfect redundancy, security, privacy or even resilience. However, this human knowledge protection system can offer an exceptional level of protection and redundancy in terms of ensuring humanity's data survives. This system will require high number of resources, coordination and maintenance to stay effective. I believe that these resources are worth it as we cannot afford to lose everything we have learnt and who we are easily. Make sure to adapt any details and protections to specific systems and resources.



The spider diagram above shows a visual layout of the human knowledge protection system. This shows the 4 layers involved and 3 features/protections of each layer. This spider diagram does not show all protections in the HKPS. More detail on protections and features can be found in the sections below.

HKPS Cloud Layer Protections and Features

- **HKPS's cloud layer is important for using existing internet infrastructure to protect and store humanity's data.**
- **The cloud layer will use a defence in depth strategy to protect itself from compromise or destruction.**
- **The first layer is self-hosted cloud storage with EZEE , zero-knowledge architecture and client-side encryption.**
- **This means that there is no third-party risk, the server/provider cannot see the data and the server does not have access to keys to the data.**
- **The second layer is integrity checking and failover systems with multiple cloud servers for redundancy.**
- **This means that the data stored in cloud storage is checked for corrupt or loss and if one cloud server were to go down the others remain.**
- **The third layer is datacentres located in private and underground areas to prevent an attacker from physically compromising it.**
- **This means that even if an attacker knows the systems exists, then they will not be able to perform any physical attack vectors without knowing where it is.**
- **The fourth layer is physical security with armed guards, keypads, reinforced doors and cameras.**
- **This means that even if an attacker knows where the datacentres are then they would have to bypass multiple layers of physical security.**
- **The fifth layer is management because each datacentre should be managed a local team of IT technicians, guards, researchers and CSIRT.**
- **This is also supported by a command structure with first, second and third in command providing redundancy.**
- **This means that an attacker must bypass an active team maintaining and protecting the data on the servers.**
- **The sixth layer is DDoS and DoS protection through load balancers, next-gen firewalls, rate-limiting and blacklisting.**
- **This means that an attacker cannot easily flood the cloud storage servers using botnets preventing data loss or data corruption.**
- **The seventh layer is endpoint protection directly for the servers through AES-256 encryption with rotating keys, automatic updates for firmware and software and logging for access attempts and sessions.**
- **This can also support by OSGuardian, RBAC, RAM design, Air-gapped when possible and with the only connection being to other servers or verified, trusted channels.**
- **You can find OSGuardian and my other designs in my security and hardware repos for more information you already haven't seen them.**
- **You may also layer more additional protections or adapt the current ones to resources and budget.**

HKPS Device Layer Protections and Features

- **HPKS's device layer is important for protecting humanity's data in its most accessible form.**
- **The device layer will also use a defence in depth strategy to protect itself from compromise or destruction.**
- **The first layer is multiple redundant workstations across different countries or different areas of a country which each store 1-2 subjects of data.**
- **This means that an attacker cannot compromise one device for all the data and it also prevents a single point of failure.**
- **The second layer is a custom operating system like a modified version of Linux with a hardened OS image for the workstations.**
- **This means that an attacker cannot use known operating system exploits that are usually seen in Windows or Mac. This also eliminates third-party risk.**
- **The third layer is physical access protections through FDE, OSGuardian, tampering detection and device tracking.**
- **This layer can also be supported by BIOS supervisor password, rate-limiting, progressive lockout, encrypted containers, 7-zip folders and time-limited sessions.**
- **This means that an attacker must compromise multiple layers protecting against physical access without being detected which raises the cost significantly.**
- **The fourth layer is custom hardware using my RAM design's encryption mode, cache, GPU and SSD design.**
- **This can also be supported by a TPM and other hardware components with no physical reset switch.**
- **This means that an attacker cannot use cold boot attacks and must bypass multiple encrypted patriations to gain access.**
- **The fifth layer is OPSEC and isolation which can be done by keeping the device air-gapped most of the time and dedicating the device specifically for storing and protecting humanity's data.**
- **This means that an attacker cannot exploit certain applications or software on the device as it isn't actively used for anything else and remote digital attacks are nearly impossible due to the air-gapped nature.**
- **The sixth layer is having physical prevention and redundancy which can be done through physical lock on the device, storing the device in a locked room and having 2 replacement devices secure and ready in case.**
- **This means that an attacker must actively bypass multiple protections before they can even boot up the device itself and it also prevents an single point of failure.**
- **You may also layer more additional protections or adapt the current ones to resources and budget.**

HKPS Physical Layer Protections and Features

- **HPKS's Physical layer is important for protecting humanity's data in one of its most vulnerable forms.**
- **The Physical layer will also use a defence in depth strategy to protect itself from compromise or destruction.**
- **The first layer is geographic redundancy and protections using multiple undergrounds vaults with reinforced doors, walls and materials.**
- **This means that an attacker would have to compromise multiple independent and separate vaults across the world or across a nation to destroy humanity's data.**
- **The second layer is secure physical access through keypads that require PIN, biometrics and keycard, clearance levels, physical searches by armed guards and cameras with real-time monitoring.**
- **This is also supported by multiple defensive positions (Sandbags, machine gun positions, towers) and armoured mounted vehicles.**
- **This means that an attacker would need to gain a clearance level, multiple physical authentication methods and bypass physical searches by guards.**
- **The third layer is liveable areas within these vaults through kitchens, bedrooms, entertainment areas, food production systems and bathrooms,**
- **This means that an attacker would need to bypass an active physical force that is rested, comfortable and prepared for physical attacks.**
- **The fourth layer is OPSEC and privacy through self-hosted software, private infrastructure, honeypots, encrypted communications and classified documents.**
- **This can also be supported through a need-to-know policy and have each team manage themselves locally to reduce the change to leaks along with intelligence agencies monitoring chatter online.**
- **This means that an attacker must bypass multiple OPSEC layers, multiple individuals and avoid intelligence agency monitoring.**
- **The fifth layer is nuclear protection systems through interception systems, nuclear-shielding and deep underground levels and honeypot signals to confuse nuclear targeting systems.**
- **This means an attacker must bypass multiple defensive systems to get a nuclear weapon through successfully which raises the cost heavily.**
- **The sixth layer is EMP protections and Storage conditions which is done by storing physical drives (SSD, HDD and Microfilm) in EMP-shielded rooms and Faraday bags, using separate rooms adding redundancy.**
- **This layer can also be supported by FDE for the drives, passwords stored in a self-hosted password manager offline or local if possible.**
- **There should also be laptops/workstations ready to ready the data when needed.**
- **You may also layer more additional protections or adapt the current ones to resources and budget.**

HKPS Human Layer Features and Protections

- **HPKS's Human Layer is important to protect the people behind the knowledge and to reduce the risk of human error.**
- **The Human layer will also use a defence in depth strategy to protect itself from compromise or destruction.**
- **The first layer is staff training which can be done through my phishing detection system, OPSEC training, security awareness training and rotation schedules.**
- **This means that attackers must bypass trained staff able to make decisions under stress and pressure without panicking reducing success rates of phishing.**
- **This also reduces the effectiveness of spear-phishing as trained staff can avoid clicking on links and reduce the amount of information an attacker can use.**
- **The second layer is fire-prevention systems through water sprinklers, fire extinguishers, smoke alarms and cooling systems for data centres and underground vaults.**
- **This means that the staff members working in the underground vaults or the datacentres will be protected from being harmed in the fire while also protecting humanity's data from being lost.**
- **The third layer is redundant power systems which can be done using backup generators stored in separate rooms and a combination of fossil fuels and green energy where possible.**
- **This can also be supported by non-generator sources such as flashlights, candles and night vision goggles.**
- **This means that staff members can work through power outages and navigate through the building or vault easily even if an attacker managed to compromise the multiple generators.**
- **The fourth layer is contingency planning and recovery planning which can be done through defined but adaptive protocols for certain situations allowing staff to deal with different incidents and keep safe.**
- **This layer can be sorted by my protocols and contingency planning structure in my security architecture. These can be found in my security repo.**
- **This means that staff members and this system is proactive and reactive instead of just reacting to threats when they arrive which can lead to mistakes and an attacker compromising more of the system.**
- **The fifth layer is final resilience protocols which can be done by using radios for communication, local networks using switches if global internet is down and backup escape routes and self-destruct system (Manual and optional)**
- **This means that staff members can still communicate in the bunker using hardware or without it while also being able to escape using multiple routes in case of complete physical compromise.**
- **It also means that an attacker does not get to access humanity's data as it can be destroyed via the self-destruct.**
- **You may also layer additional protections or adapt the existing ones.**

HKPS Daily Workflow

- This workflow will show what an average day of this system may look.
- 6:00 AM: Night shift to morning shift changes happens. Morning shift guards take up defensive positions, watch camera footage and patrol down defined routes.
- 7:00 – 8:30 AM: All Datacentres, undergrounds vaults and device storage areas confirmed all clear signal to each other no alerts and Kitchen Staff begin preparing food
- 8:30 AM: All other staff members are woken up and called for roll call before breakfast to ensure that all staff members are there.
- 8:40 → 9:15 AM: Staff members eat breakfast and given time to wake up before being given their daily tasks.
- 9:15 AM: All staff members are given their tasks for the day. CSIRT reviews security logs. IT teams check performance logs. Engineers check on generators.
- 9:30 AM: External staff members arrive to the facility. Guards perform physical checks for any weapons or any tracking equipment. External staff members use keycards to enter the facility.
- 10:00 AM: Tamper detection workstations run and IT teams check data for corruption or loss.
- 11:00 AM : Intelligence teams confirm that backup exit routes are clear and aren't being monitored by an attacker and Kitchen Staff begin preparing for lunch.
- 12:00 PM: Morning shift to afternoon shift change happens. Afternoon shift guards take up defensive positions, watch camera footage and patrol down defined routes.
- 1:00-1:45 PM: Staff members have lunch, time to rest or head to one of the rooms to enjoy something fun.
- 2:00 PM: A Password Review and Keycard check is performed on the offline password manager for each staff member by IT team and CSIRT.
- 4:00 PM: A fire drill is performed where staff follow planned protocols, use emergency exits and do a headcount once clear.
- 6:00 PM: All Datacentres, underground vaults and device storage areas confirm all clear signal to each other with no alerts and Kitchen staff prepare for dinner.
- 7:30- 8:00 PM: Staff members are given time to eat, rest and enjoy themselves in the facility.
- 8:15 PM: External staff members leave the facility and are checked by guards for any physical drive or important equipment from the facility to ensure that staff haven't stolen anything.
- 8:30 PM: Afternoon shift guards swap with night shift guards as night shift guards take up defensive positions, watch camera footage and take patrols.
- 12:00 AM: integrity check ins performed on servers and workstations and a full network backup is performed

HKPS Trade-Offs

- The human knowledge protection system also has trade-offs just like any other system I have built.
- It is important to remember that there may be additional trade-offs during actual implementation
- I have identified 6 main trade-offs for the HKPS.
- The first trade-off is cost.
- This trade-off exists as global infrastructure, maintenance and staff wages can be an extremely massive budget situation.
- This trade-off can be managed by budget management, national handling and efficient coding and infrastructure practices.
- The second trade-off is management load.
- This trade-off exists as managing multiple datacentres, vaults and staff members can be extremely difficult and exhaustive.
- This trade-off can be managed through local command structures, national handling of datacentres and vaults and team organisation for global management.
- The third trade-off is cognitive load
- This trade-off exists as staff members and intelligence teams can get exhausted from following a disciplined routine daily and being alert 24/7.
- This means that if staff members are under too much stress and pressure, they can be more likely to make mistakes which can compromise.
- This trade-off can be managed by rotating schedules, rest periods, providence of water & food and available emotional support.
- The fourth trade-off is performance.
- This trade-off exists as all the security features and redundancy features can be computationally intensive which can slow down the workstations.
- This means that staff members will not be able to respond to incidents or breaches quickly if devices are slow.
- This trade-off can be managed through network management and custom hardware using my designs or other designs if needed.
- The fifth trade-off is time.
- This trade-off exists because building global infrastructure, coordination and organisational can take years to complete.
- This trade-off can be managed through structured development processes, nation-specific efforts and using pre-existing coordination like the UN or EU.
- The sixth trade-off is human error
- This trade-off exists because humans can be stressed, exhausted, scared or sad which means there can be security errors.
- This trade-off can be managed through training and technical controls such as RBAC, AV and zero-knowledge architecture as a safety net.